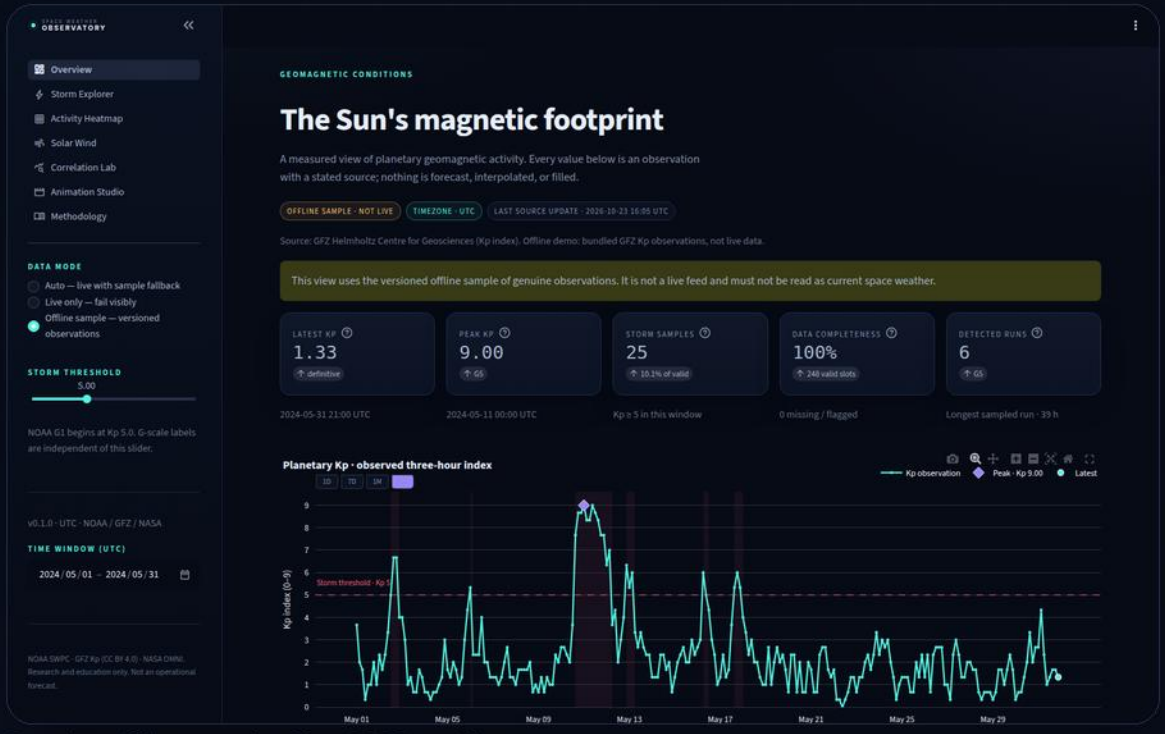
WHEN THE SUN DISRUPTS EARTH

When the Sun disrupts Earth.

An evidence-first observatory for geomagnetic storms: real observations, explicit uncertainty, and no zero-fill.

NOAA / GFZ / NASA · 58 TESTS · PYTHON · STREAMLIT



Genuine offline sample · May 2024 · not live

github.com/Shahbaz2104/space-weather-observatory

Three ways a chart can lie.

None of them require bad data. They only require a misleading default.

01 **A gap drawn as quiet**

Missing observations become a flat line.

02 **An interpolated curve drawn as measurement**

Smoothing invents values that were never observed.

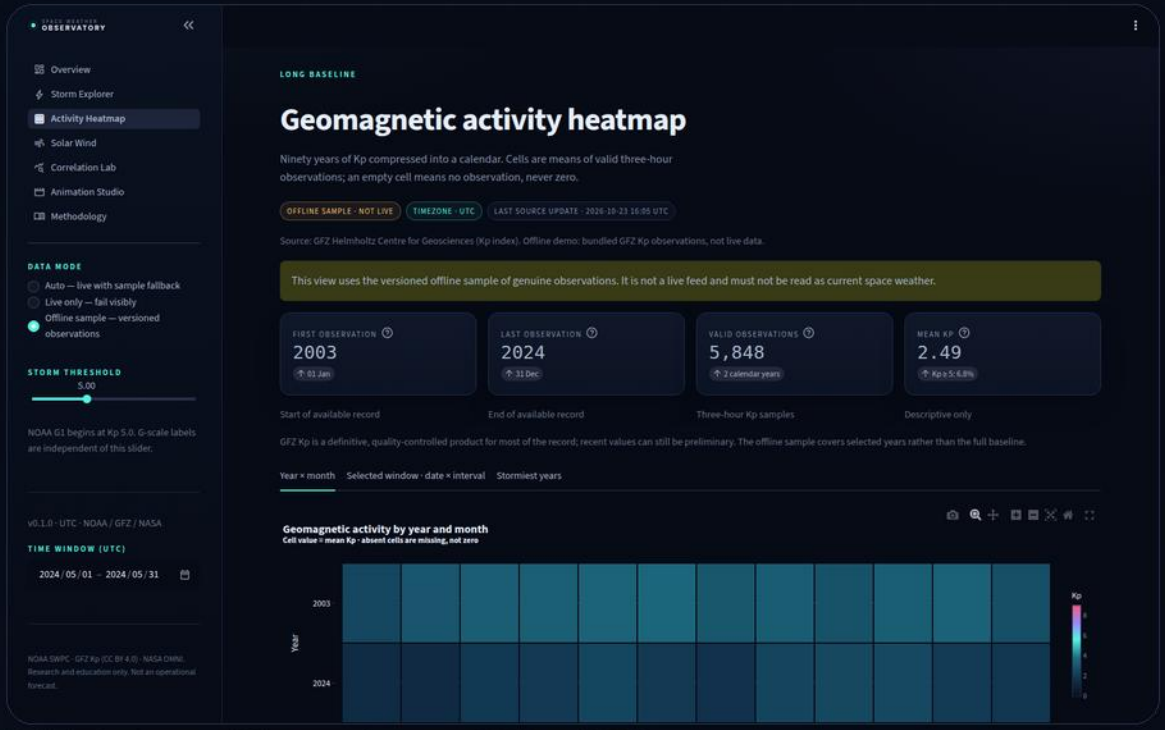
03 **A Kp value confused with a storm**

An activity index is not an event catalogue.

Seven linked views. One time convention.

From a live Kp overview to a ninety-year heatmap, every screen uses UTC and keeps missing observations missing.

Overview · Storm Explorer · Activity Heatmap · Solar Wind · Correlation Lab · Animation Studio · Methodology



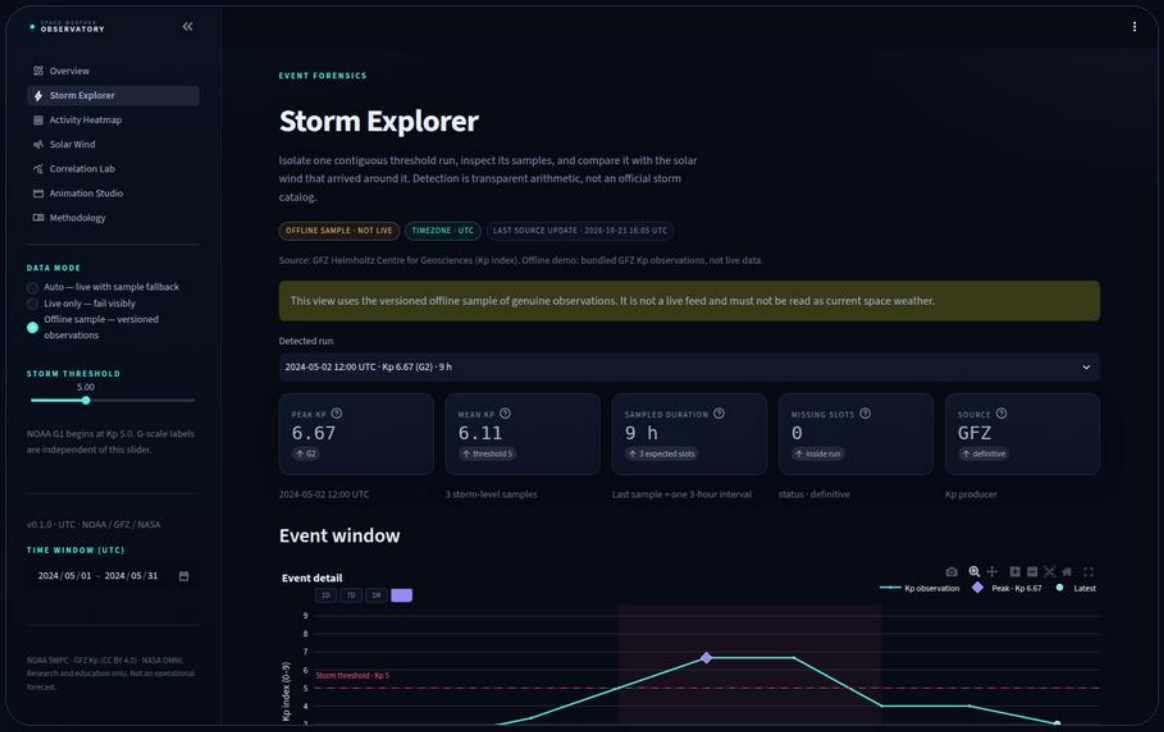
Activity heatmap · offline sample covers 2003 and 2024

github.com/Shahbaz2104/space-weather-observatory

Missing stays missing.

No interpolation. No zero-fill. No silent gap closure.

Display sampling keeps endpoints and missing-run boundaries, so a long series cannot be made to look continuous.



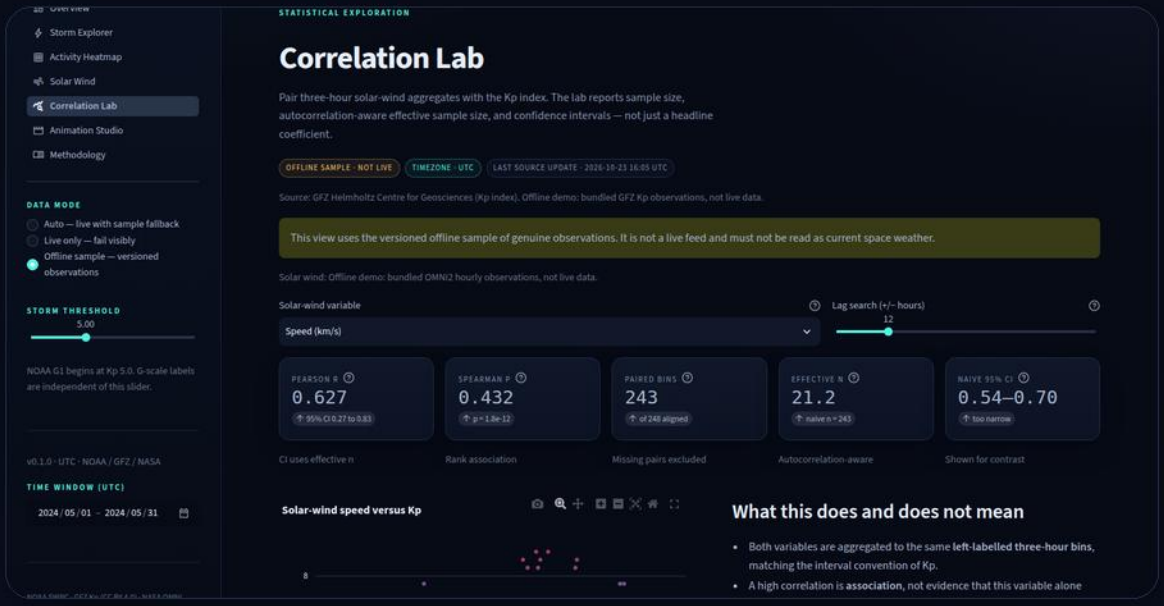
Storm Explorer · sampled span, missing slots, and provenance

JSON provenance is available for exports: sources, timestamps, units, transformations, threshold, and warnings.

A correlation is not a mechanism.

The lab reports the sample size and autocorrelation-aware uncertainty, not just the headline coefficient.

PEARSON r	EFFECTIVE n	PAIRED BINS
0.627	21.2	243
95% CI uses effective n	autocorrelation-aware	missing pairs excluded



May 2024 offline sample · solar-wind speed versus Kp

INSPECT THE EVIDENCE

Seven pages. Real observations.

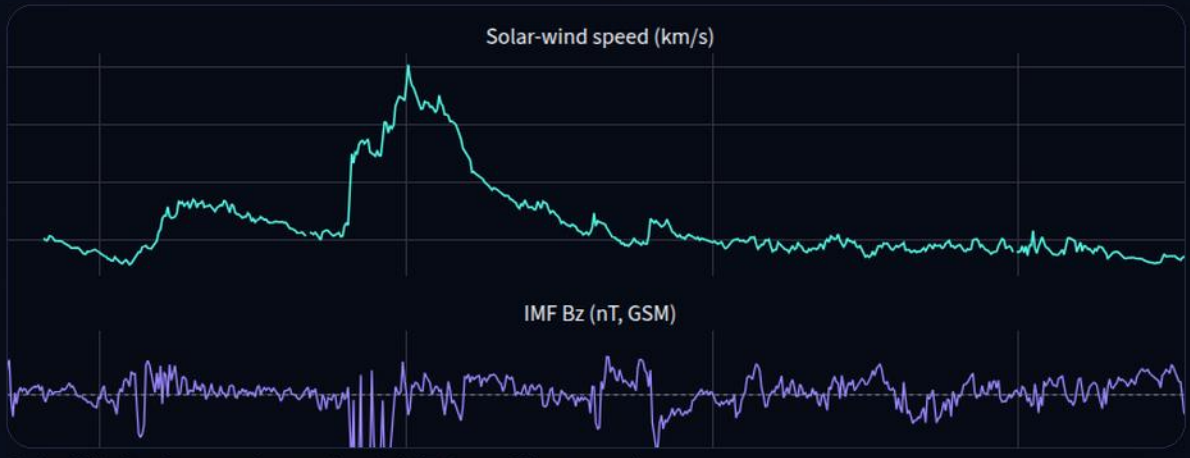
Python, Streamlit, Plotly, pandas, NumPy, SciPy, Matplotlib. Built to make the caveats as legible as the numbers.

LIVE DEMO
https://space-weather-observatory-nsnhccmkm2xumzqopwzi7q.streamlit.app/overview

REPOSITORY
github.com/Shahbaz2104/space-weather-observatory

Live mode and the offline sample are labelled. Not a forecast service.

58 TESTS · RUFF · MYPY | GITHUB ACTIONS | MIT LICENSED CODE



Solar Wind · observed speed and IMF Bz · offline sample